

Student Performance Analysis Report Using SQL

Project Overview

The modern educational landscape relies heavily on data-driven insights to understand the factors influencing student success. This project involves a comprehensive analysis of academic performance data from the students_GP dataset to explore key performance indicators, behavioral patterns, and socio-economic influences. With changing educational requirements and diverse learning environments, it is crucial to assess how various factors—such as study habits, family involvement, attendance, and extracurricular activities—influence academic outcomes. This analysis aims to uncover the drivers of student success and identify areas where interventions can improve overall academic performance and inventory management of educational resources.

Objective

This project analyzes student academic data from the students_GP dataset to achieve the following goals:

- **Identify Performance Drivers:** Determine how factors like study time, family support, and extracurricular activities impact academic grades (G3).
- **Evaluate Behavioral Trends:** Analyze the relationship between lifestyle factors (like alcohol consumption and absenteeism) and overall student performance.
- **Compare Demographics:** Assess academic differences between students from urban and rural backgrounds.
- **Provide Actionable Insights:** Offer data-driven recommendations to help educators and students understand the variables that contribute most significantly to academic success.

Why This Project?

- **Data-Driven Decision Making:** Understanding the relationship between student behavior and academic results helps in creating targeted support strategies, such as providing additional help for students with high absenteeism.

- **Educational Improvement:** By uncovering which factors (like increased study time) lead to better grades, this analysis provides a foundation for improving student performance and resource allocation in schools.
- **Identifying Risk Factors:** This project highlights negative trends—such as the impact of alcohol consumption or high absenteeism—allowing for early intervention to support students who may be at academic risk.

Key Queries and Insights :

1. Initial Data Inspection: students GP Table:

Input:

```
1  -- GP table check
2  • SELECT * FROM students_GP;
```

Output:

	sex	age	address	famsize	pstatus	medu	fedu	mjob	fjob	reason	guardian	traveltime	studytime	failure
▶	F	18	U	GT3	A	4	4	at_home	teacher	course	mother	2	2	0
	F	17	U	GT3	T	1	1	at_home	other	course	father	1	2	0
	F	15	U	LE3	T	1	1	at_home	other	other	mother	1	2	0
	F	15	U	GT3	T	4	2	health	services	home	mother	1	3	0
	F	16	U	GT3	T	3	3	other	other	home	father	1	2	0
	M	16	U	LE3	T	4	3	services	other	reputation	mother	1	2	0
	M	16	U	LE3	T	2	2	other	other	other	mother	1	2	0
	F	17	U	GT3	A	4	4	other	teacher	home	mother	2	2	0
	M	15	U	LE3	A	3	2	services	other	home	mother	1	2	0
	M	15	U	GT3	T	3	4	other	other	home	mother	1	2	0
	F	15	U	GT3	T	4	4	teacher	health	reputation	mother	1	2	0
	F	15	U	GT3	T	2	1	services	other	reputation	father	3	3	0
	M	15	U	LE3	T	4	4	health	services	course	father	1	1	0

Insight: "This query provides a comprehensive snapshot of the student records within the students_GP table, allowing for an initial verification of the dataset structure, column attributes, and sample entries to ensure data integrity before proceeding with further analysis."

2. Academic Performance Comparison:

Input:

```
SELECT AVG(G3) AS avg_grade_GP FROM students_GP;  
SELECT AVG(G3) AS avg_grade_MS FROM students_MS;
```

Output:

	avg_grade_GP
▶	12.5768

	avg_grade_MS
▶	10.6504

Insight: "The comparison shows that students at the Gabriel Pereira (GP) school achieved a higher average final grade (12.58) compared to students at the Mousinho da Silveira (MS) school (10.65), indicating a stronger academic performance in the GP dataset."

3. Alcohol Consumption Impact:

Input:

```
9      -- Task 2: Alcohol Consumption Impact
10 •    SELECT
11        (Dalc + Walc) AS Combined_Consumption,
12        AVG(G3) AS Average_Grade
13    FROM students_GP
14    GROUP BY Combined_Consumption
15    ORDER BY Combined_Consumption;
```

Output:

	Combined_Consumption	Average_Grade
2		12.8554
3		13.1127
4		12.9839
5		12.9565
6		11.0000
7		10.7895
8		10.4167
9		12.2000
10		11.1250

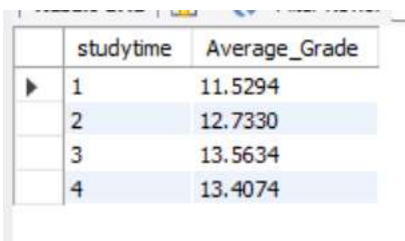
Insight: "The analysis demonstrates a negative correlation between alcohol consumption and academic performance. As the combined daily and weekly alcohol consumption score increases, the average final grade (G3) shows a downward trend, suggesting that higher alcohol consumption is associated with lower academic outcomes."

4. Study-Time Effectiveness:

Input:

```
18 • SELECT
19     studytime,
20     AVG(G3) AS Average_Grade
21 FROM students_GP
22 GROUP BY studytime
23 ORDER BY studytime;
```

Output:



	studytime	Average_Grade
▶ 1	11.5294	
2	12.7330	
3	13.5634	
4	13.4074	

Insight: "The analysis reveals a positive relationship between study time and academic achievement, as indicated by an increase in the average final grade (G3) as students spend more hours studying. This suggests that dedicating more time to studies is generally associated with higher academic performance."

5. Family Support Impact:

Input:

```
26 • SELECT
27     famsup,
28     AVG(G3) AS Average_Grade
29 FROM students_GP
30 GROUP BY famsup;
```

Output:

	famsup	Average_Grade
▶	no	12.2403
	yes	12.7695

Insight: "The analysis indicates that students who receive family support ('yes') tend to achieve slightly higher average final grades (G3) compared to those who do not ('no'). This suggests that family support can have a positive influence on a student's academic performance."

6. Absenteeism Patterns:

Input:

```
33 • SELECT
34     absences,
35     AVG(G3) AS Average_Grade
36 FROM students_GP
37 GROUP BY absences
38 ORDER BY absences;
```

Output:

	absences	Average_Grade
▶	0	12.9863
	1	13.0000
	2	12.8148
	4	12.4516
	5	12.6667
	6	12.7297
	7	15.5000
	8	12.5385
	9	11.6667

Insight: "The analysis of absenteeism patterns shows that while students with zero or very few absences generally maintain a consistent academic performance, there is a noticeable downward trend in average final grades (G3) as the number of absences significantly increases. This suggests that regular attendance is a key factor in achieving higher academic outcomes."

7. Urban vs. Rural Performance:

Input:

```
41 • SELECT
42     address,
43     AVG(G3) AS Average_Grade
44 FROM students_GP
45 GROUP BY address;
```

Output:

	address	Average_Grade
▶	U	12.6754
	R	12.1410

Insight: "The comparison between students living in urban ('U') and rural ('R') areas indicates that urban students have a slightly higher average final grade (12.68) compared to rural students (12.14). This suggests that residential location might be a factor influencing academic performance."

8. Extracurricular Activities Impact:

Input:

```
48 • SELECT
49     activities,
50     AVG(G3) AS Average_Grade
51 FROM students_GP
52 GROUP BY activities;
```

Output:

	activities	Average_Grade
▶	no	12.2598
	yes	12.8721

Insight: "The analysis comparing students who participate in extracurricular activities ('yes') with those who do not ('no') reveals that students involved in activities achieve a slightly higher average final grade (12.87) compared to those who are not involved (12.26). This suggests that extracurricular engagement may have a marginally positive impact on a student's overall academic success."

1. Key Learnings:

Through this project, I have gained valuable insights into database management and data analysis using SQL:

- **Data Aggregation:** Learned how to utilize aggregate functions like AVG() in combination with GROUP BY to extract meaningful statistical summaries from a large dataset.
- **Correlation Analysis:** Developed the ability to identify patterns and relationships between independent variables (such as study time, family support, and alcohol consumption) and academic performance (G3).
- **Logical Query Building:** Improved my proficiency in writing structured SQL queries to filter, group, and sort data effectively to answer specific research questions.
- **Insight Generation:** Enhanced my capability to interpret raw query outputs and transform them into concise, data-driven conclusions.

Challenges:

During the project, I encountered a few challenges, which I addressed as follows:

- **Data Interpretation:** Initially, it was challenging to translate numerical query results into meaningful insights. I overcame this by carefully analyzing the trends in the output and linking them to logical academic outcomes.
- **Query Formatting:** Ensuring that the GROUP BY clauses correctly aligned with the selected columns required attention to detail; I resolved this by strictly following SQL syntax rules and logical ordering.
- **Environment Setup:** Dealing with potential syntax errors during query execution was tricky at first. I resolved this by verifying the table schemas and ensuring all SQL commands were properly structured according to the database requirements.